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Project Title:	Orientational ordering, pair interactions, and controlled self-assembly of magnetic nanowires in nematic liquid crystal solvents		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Liquid crystals		
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Task Description:

The primary objective of this ground-based research program is the experimental study of orientational ordering, pair interactions, and controlled self-assembly of magnetic nanowires suspended in liquid crystals. A colloidal particle in a liquid crystal matrix can introduce defects and other distortions in the liquid crystal's ordered state due to the boundary conditions created by the anchoring of the liquid crystal molecules at the particle’s surface. These distortions can, in turn, orient aspherical colloids with respect to the liquid crystal as well as mediate anisotropic interactions between particles. Experiments are proposed to characterize the orientational energy of wire-shaped colloids and their pair interactions within nematic liquid crystals for a variety of anchoring conditions. The use of magnetic particles will allow controlled studies that include the application of torques and forces to the wires through external fields and field gradients that compete with the elastic forces of the nematic. Since the magnetic nanowires are potential building blocks for nanostructured devices, a particular focus of the work will be toward creating conditions for self-assembly of the wires into ordered configurations that would facilitate the study of their electronic and magnetic properties or enhance their applications potential. The specific hypotheses driving the research strategy are: (i) close investigation of the nematic distortions around suspended wire-shaped colloids and the resulting interactions among the colloids can provide insightful perspectives on inclusions in liquid crystals, including the generation of defects; (ii) through judicious choice of surface treatments, the wire-wire interactions mediated by nematic distortions can be tuned to influence wire self-assembly; (iii) distortions in the nematic director field can be tailored to manipulate suspended wires leading to complex assemblies with useful properties; (iv) efforts to counter gravity can affect nanowire behavior, creating to new opportunities for fundamental study and application. To test these hypotheses, the PI's will build on their combined expertise in liquid crystal physics, magnetism, electrodeposition for designing and fabricating magnetic nanowires, chemical techniques for functionalizing wire surfaces, and video microscopy for characterizing the behavior of suspended wires and the surrounding nematic. Because of the inherent density mismatch between the metal wires and the liquid crystal solvent, gravity strongly impacts the behavior of suspensions, and experiments are proposed to illuminate its effects. The resulting understanding of the role of gravity in the orientational ordering and self-assembly of magnetic nanowires in liquid crystals will provide a basis for the future development of space-based studies that can take advantage of NASA's expanding fluid physics capabilities. This research task has several specific objectives:
Objective 1: to determine the orientational forces imposed on isolated nanowires suspended in a nematic liquid crystal.

Objective 2: to characterize the dynamical behavior of isolated nanowires suspended in a nematic liquid crystal in response to controlled forces and torques.

Objective 3: to determine the interactions between pairs of nanowires and between nanowire and other objects mediated by nematic distortions.

Objective 4: to develop methods for the manipulation of suspendend nanowires self-assembly that utilize the unique interactions experienced by nanowires in liquid crystals.

Objective 5: to develop methods for the self-assembly of nanowires that utilize the unique interactions experienced by nanowires in liquid crystals.

Objective 6: to modify surface properties of metal nanowires in order to control their nematic anchoring properties.

Objective 7: to investigate the effects of gravitational forces on nanowire interactions and self-assembly.

Objective 8: to develop instrumental tools that can mimic gravitational effects, and in particular can create effective low-gravity environments, for microscopy studies of magnetic nanowires in fluid suspension. Colloids suspended in a complex fluid can be subject to a variety of interactions including electromagnetic, entropic, and gravitational forces. These interactions form the basis for many important topics in science and technology; two examples are the emergent field of microrheology and the self-assembly of colloidal particles. For colloids in a liquid crystal, the fluid’s elastic energy provides a unique mechanism for interactions. Our research seeks to make significant contributions to the field of colloids by pursuing the following hypotheses: (i) close investigation of the nematic distortions around suspended wire-shaped colloids and the resulting forces on the colloids can provide insightful perspectives into liquid crystals, including the generation of defects; (ii) through judicious choice of surface treatments, the wire-wire interactions mediated by nematic distortions can be tuned to influence wire self-assembly; (iii) distortions in the nematic director field can be tailored to manipulate suspended wires, leading to complex assemblies with useful properties; and (iv) efforts to counter gravity can both affect nanowire behavior and lead to instrumentation development that impacts the study and application of magnetic colloids in fluids in general. Further, since the magnetic nanowires represent potential building blocks for nanostructured devices, particularly in magnetoelectronics, our ability to create conditions for self-assembly into ordered configurations will facilitate the study of their electronic and magnetic properties in ways that can enhance their utility in such applications.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research effort during year 1 of the grant has concentrated on four topics.
A) Orientational elastic energy of nanowires with planar anchoring in a nematic solvent.

We have measured the elastic torque on a colloidal wire as a function of angle theta between the wire's long axis and the nematic director. These measurements were performed by balancing the elastic torque on a nickel wire suspended in the nematic with the magnetic torque from an external field. The torque grows linearly with angle, and hence the elastic energy grows quadratically. Theory by Brochard and de Gennes, based on an analogy between the elastic energy and electrostatics, predicts that the energy U varies with theta as U = 6.28KC(theta)^2, where K is the nematic's average elastic constant, and remarkably, C is the electrical capacitance of the wire. Tracking the torque as a function of temperature, we have shown that the measured energy agrees quantiatively with this prediction.

B) Angular relaxation of nanowires with planar anchoring in a nematic solvent: a probe of anisotropic viscosity.

Due to the elastic torque, a wire initially at some angle with respect to the director will relax back to a parallel orientation. We have characterized the dynamics of this process and have shown quantitatively how it is depends on the anisotropic drag of the nematic fluid. Specifically, from the wire's motion at theta near 90 degrees and near 0 degrees, we identify two effective viscosities that we can relate to two of the nematic's Miesowicz coefficients. These identifications imply a somewhat unexpected orientation of shear flow relative to director field around a cylinder

Task Progress:	rotating in a nematic fluid. C) Orientation-dependent repulsive interaction with a wall. We have characterized the elastic interaction between a wire and a flat surface with planar anchoring. We find that there is an orientation-dependent repulsive force between wire and surface that we have measured quantitatively by balancing the force with gravity. The results for the elastic energy agree closely with calculations based on an extension of the electrostatic analogy developed by Brochard and de Gennes to this experimental geometry. D) Manipulating wires with director field gradients: A demonstration experiment using a twisted nematic. We have performed experiments to demonstrate an approach to manipulating nanowires through a combination of external magnetic fields and appropriately designed director-field configurations. The measurements involved a twisted nematic cell. In a twisted cell the liquid crystal is contained between parallel glass slides with rubbed polymer surfaces that anchor the director parallel to each surface. The surfaces are oriented such that the two anchoring directions are perpendicular to one another. As a result, in going from the bottom surface to the top, the director rotates with a uniform pitch. We have shown that, due to the elastic energy, a wire in a twisted cell that is oriented at a fixed angle by an external magnetic field feels an elastic force that drives it to the height where the local director orientation matches that of the wire. We have employed this technique to achieve sub-micrometer positioning of suspended magnetic nanowires.
Bibliography Type:	Description: (Last Updated: 04/30/2007)
Articles in Peer-reviewed Journals	Lapointe, C; Cappallo, N.; Reich, D. H.; Leheny, R. L. "Static and Dynamic Properties of Magnetic Nanowires in Nematic Fluids" Journal of Applied Physics, in press , May-2005
Presentation	Leheny, R. L. "Anisotropic Particles within Anisotropic Fluids" H. G. Herb Materials Physics Seminar, University of Wisconsin, Madison, WI Apr-2004
Presentation	Leheny, R. L. "Anisotropic Particles within Anisotropic Fluids" Condensed Matter Seminar, University of Massachusetts, Amherst, MA Sep-2004
Presentation	Lapointe, C. "Elastic interactions and the manipulation of colloidal wires in a nematic solvent" International Liquid Crystal Conference, Ljubljana, Slovenia Jul-2004
Presentation	Lapointe, C. "Elastic interactions and the manipulation of colloidal wires in a nematic solvent" American Chemical Society National Meeting, Philadelphia, PA Aug-2004
Presentation	Leheny, R L. "Anisotropic Particles within Anisotropic Fluids" W. G. Pritchard Seminar, Pennsylvania State University, University Park, PA Sep-2004
Presentation	Leheny, R. L. "Elastic Torque and the Manipulation of Magnetic Nanowires in Anisotropic Complex Fluids" 49th Conference on Magnetism and Magnetic Materials, Jacksonville, FL Nov-2004
Presentation	Leheny, R. L. "Anisotropic Particles within Anisotropic Fluids" NIST Center for Neutron Research, Gaithersburg, MD Jan-2005
Presentation	Cappallo, N. "Formation of localized shear induced states in wormlike micelle solutions using magnetic nanowire probes" 2005 March Meeting of the American Physical Society Mar-2005